From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

PART TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 210**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

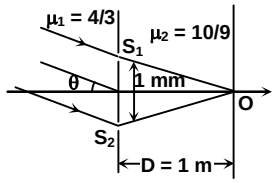
PART – I

SECTION – A

Single Correct Choice Type

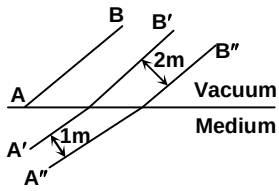
This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In YDSE experiment shown in the figure, a parallel beam of light of wavelength (λ = 0.3 mm) in medium μ_1 is incident at an angle $\theta = 30°$ as shown. ($S_1O = S_2O$). If the intensity due to each light wave at point O is I_0 then the resultant intensity at point O will be



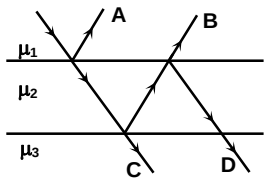
(A) zero
(B) $2I_0[1 + \cos(40\pi/9)]$
(C) $3I_0$
(D) I_0

2. Plane wavefronts are incident on a medium as shown in the figure. The refractive index of the medium is



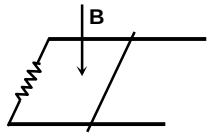
(A) 2
(B) 4
(C) 1.5
(D) can't be predicted.

3. A ray of light is incident on a thin film as shown in the figure. A and B are two reflected rays from first surface and C & D are two final transmitted rays. Rays B and D undergo a phase change π. Choose the correct relation among refractive indices is



(A) $\mu_2 > \mu_3 > \mu_1$
(B) $\mu_3 > \mu_2 > \mu_1$
(C) $\mu_3 > \mu_1 > \mu_2$
(D) None of these

4. Two parallel rails of negligible resistances are fixed on a horizontal surface. One end of the rails are connected by a resistance R as shown in the figure. There is a uniform magnetic field B perpendicular to the plane of the rails. A sliding wire of negligible resistance, length ℓ and mass m can slide on the rails. The coefficient of friction between the rails and wire is μ. The minimum force required to move the wire with uniform velocity v is



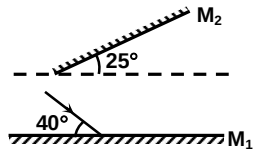
(A) $\mu mg + \frac{B^2a^2}{R}v$
(B) $\frac{B^2a^2}{R}v$
(C) $\frac{\mu mg}{\sqrt{\mu^2+1}} + \frac{B^2a^2}{R}\frac{v}{\sqrt{\mu^2+1}}$
(D) None of these

Space for rough work

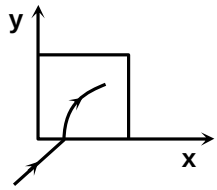
5. The minimum angle of deviation of a prism of refractive index μ is equal to its refracting angle. The refracting angle of prism is
(A) $2\cot^{-1}(\mu/2)$ (B) $2\tan^{-1}(\mu/2)$
(C) $2\sin^{-1}(\mu/2)$ (D) $2\cos^{-1}(\mu/2)$

6. A ray of light moving along the vector $\left(\hat{i}-2\hat{j}\right)$ undergoes refraction at an interface of two media, which is x-z plane. The refractive index for y > 0 is 2 while for y < 0 it is $\sqrt{5}/2$. The unit vector along the refracted ray is
(A) $\dfrac{-3\hat{i}-5\hat{j}}{\sqrt{34}}$ (B) $\dfrac{-\left(4\hat{i}-5\hat{j}\right)}{5}$
(C) $\dfrac{-3\hat{i}-4\hat{j}}{5}$ (D) $\dfrac{4\hat{i}-3\hat{j}}{5}$

7. By what angle should M_2 be rotated, so that the light ray after reflection from both the mirrors become horizontal
(A) 95° clockwise (B) 75° anticlockwise
(C) 5° anticlockwise (D) 15° anticlockwise

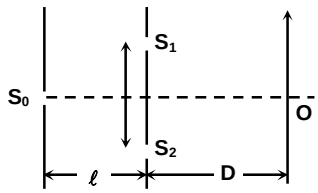


8. A ray of light is incident on a medium having variable refractive index. In which of the following variation of μ will the ray follow the path shown in the figure.
(A) $\mu = ax + b$ (a, b are positive)
(B) $\mu = a - bx$ (a, b are positive)
(C) $\mu = ay + b$ (a, b are positive)
(D) $\mu = a + by^2$ (a, b are positive)



9. A plano convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered on its flat surface. Where should an object be placed relative to the silvered lens so as to form a real image of the same size as the object?
(A) 10 cm (B) 20 cm
(C) 30 cm (D) 60 cm

10. The figure shows a schematic diagram for Young's double slit experiment. Given $d \ll \ell$, $d \ll D$, $\lambda/D \ll 1$. Which of the following is a right statement about the wavelength of light used?
(A) Larger the wavelength, larger will be the fringe width.
(B) If white light is used, violet colour forms its first maxima closest to the central maxima.
(C) The central maxima of all wavelength coincide.
(D) All of the above

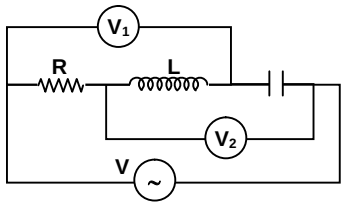


Space for rough work

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

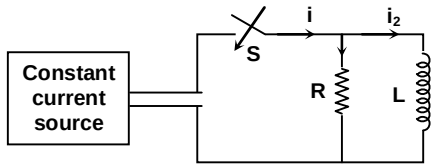
11. In the RLC series circuit shown, the reading of voltmeters are V_1 = 150 V, V_2 = 50 V and the source has emf 130 V.
(A) power factor of circuit will be 12/13
(B) value of V_L = 90 V
(C) value of V_C = 40 V
(D) value of V_L = 50 V



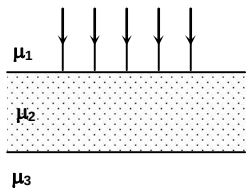
12. In β-decay, the Q value of the process is E. Then
(A) K.E. of β-particle cannot exceed E
(B) K.E. of antineutrino emitted lies between zero and E
(C) N/Z ratio of the nucleus is altered
(D) mass number (A) of the nucleus is altered

13. If the potential energy is taken zero at ground state of hydrogen atom in place of infinity then
(A) total energy of atom will change
(B) Binding energy of atom in ground state will remain unchanged
(C) kinetic energy of electron in ground state will get changed
(D) Energy of photon generated because of de excitation of atom from first excited state will remain unchanged

14. In the circuit shown switch S is close at t = 0. There after the constant current source by varying its emf maintains a constant current i out of its upper terminal. Then which of following are/is correct.
(A) Current at t = 0 is maximum from resistor and it decreases to zero after long time
(B) Current at t = 0 is zero and it increases to maximum value.
(C) Rate of growth of current in inductor is equal to rate of decay of current in inductor
(D) Current in inductor and resistor are equal at instant t = $\frac{L}{R}\ln 2$



15. Consider a case of thin film interference as show. Thickness of film is equal to wavelength of light in μ_2.
(A) Reflected light will be maxima if $\mu_1 < \mu_2 < \mu_3$
(B) Reflected light will be maxima if $\mu_1 < \mu_2 > \mu_3$
(C) Transmitted light will be maxima if $\mu_1 > \mu_2 > \mu_3$
(D) Transmitted light will be maxima if $\mu_1 > \mu_2 < \mu_3$



Space for rough work

SECTION –C

Integer Answer Type

This section contains **5 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. A non conducting ring of mass m = 1 kg and radius R = 10 cm has a charge Q = 2 C uniformly distributed over its circumference. The ring is placed on rough horizontal surface such that plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0t^2$ where $B_0 = 4T/sec^2$ is switched on. After 5 sec, after switching on the magnetic field the ring is just about to rotate about vertical axis through the centre. The coefficient friction μ between the ring and the surface is found to be $x \times 10^{-}$. Find the value of x.

2. A solenoid has an inductance of 10 H and resistance $R = 5\Omega$, it is connected to 10 V battery. It take for the magnetic energy to reach ¼ of its maximum value is xln2. Then value of x is

3. A thin prism P_1 of angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 so that ray undergoes no diviation. The angle of prism P_2 in degree is

4. Young's double slit experiment is performed with light consisting of two wavelengths $\lambda_1 = 6000Å$ and $\lambda_2 = 4800Å$. It is observed that the maximum of the 16th order corresponding to $\lambda_1 = 6000$ Å coincides with the maximum of the nth order corresponding to $\lambda_2 = 4800$ Å. The value of n is 10K. Find K.

5. Find the effective value or rms value of an alternating current (in ampere) that changes according to the law (all quantities are in S.I. unit and symbols have their usual meaning

   I = 10, when $0 < t < T/8$ I = 0, when $T/8 < t < T/2$

   I = –10, when $5T/8 < t < T/2$ I = 0, when $5T/8 < t < T$

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which only one is correct.

1. Which one of the following is true for Schottky defect in crystals?
   (A) Schottky defects increases the density of the crystal.
   (B) Schottky defect is found in crystal having large $\frac{r^+}{r^-}$ ratio.
   (C) Schottky defects are non thermodynamic defects.
   (D) Schottky defect is shown by highly ionic compounds having high coordination number.

2. The number of P–O–P bonds in cyclic trimetaphosphoric acid is
   (A) Zero (B) Two
   (C) Three (D) Four

3. In the coagulation of a positive sol, the flocculation powers of Cl^-, SO_4^{2-}, PO_4^{3-} and $[Fe(CN)_6]^{4-}$ are in the order of
   (A) $Cl^- > SO_4^{2-} > [Fe(CN)_6]^{4-} > PO_4^{3-}$ (B) $Cl^- > PO_4^{3-} > SO_4^{2-} > [Fe(CN)_6]^{4-}$
   (C) $[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^-$ (D) $Cl^- > SO_4^{2-} > PO_4^{3-} > [Fe(CN)_6]^{4-}$

4. E° values of Mg^{2+}/Mg is –2.37 volts of Zn^{+2}/Zn is –0.76 volt and Fe^{+2}/Fe is –0.44 volt. Which of the following statements is correct?
   (A) Zn oxidize Fe (B) Zn will reduce Fe^{2+}
   (C) Zn will reduce Mg^{2+} (D) Mg oxidizes Fe

Rough work

5. Lowering of vapour pressure ΔP_1 of aqueous solution of acid HA_1 and ΔP_2 of aqueous solution of acid HA_2 are related as $\Delta P_1 > \Delta P_2$. Correct relation is (Assume same concentration for both acids)
(A) $pK_{a_1} > pK_{a_2}$ (B) $pK_{a_1} < pK_{a_2}$
(C) $pK_{a_1} = pK_{a_2}$ (D) none of these

6. Which equation does not involve the reduction of a transition metal compound?
(A) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$ (B) $TiO_2 + 2C + 2Cl_2 \longrightarrow TiCl_4 + 2CO$
(C) $Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$ (D) $TiCl_4 + 4Na \longrightarrow Ti + 4NaCl$

7. The aqueous solution of a certain compound (X), when added to a copper sulphate solution, a brown precipitate is obtained which turns white on addition of excess of $Na_2S_2O_3$ solution. Its aqueous solution, when added to Ag^+ ion solution, a yellow curdy precipitate is obtained which is insoluble in ammonium hydroxide. Hence, compound (X) is
(A) NaBr (B) KBr
(C) KI (D) $PbCrO_4$

8. A solution containing 0.319 g of complex $CrCl_3.6H_2O$ was passed through cation exchanger and the solution given out was neutralized by 28.5 ml of 0.125 M NaOH. What is the correct formula of complex?
(A) $[Cr(H_2O)_6]Cl_3$ (B) $[Cr(H_2O)_5Cl]Cl_2.H_2O$
(C) $[Cr(H_2O)_3Cl_3].3H_2O$ (D) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$

9. Which of the following complexes is an inner d- orbital complex?
(A) $[Co(F)_6]^{-3}$ (B) $[Fe(F)_6]^{-3}$
(C) $[Cr(NH_3)_6]^{3+}$ (D) $[Co(H_2O)_6]^{3+}$

10. Ferric ion forms a prussian blue coloured ppt. due to formation of
(A) $KMnO_4$ (B) $K_4[Fe(CN)_6]$
(C) $Fe_4[Fe(CN)_6]_3$ (D) $Fe(OH)_3$

Rough work

Multiple Correct Answers Type

This section contains **5 multiple correct** answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.

11. 1 mole benzene ($p^0_{benzene} = 42$ mm) and 2 mole toluene ($p^0_{benzene} = 36$ mm) will have:
    (A) total vapour pressure 38 mm
    (B) mole fraction of vapours of benzene above liquid mixture is $\frac{7}{19}$
    (C) mole fraction of vapours of benzene above liquid mixture is $\frac{1}{3}$
    (D) mole fraction of vapours of toluene above liquid mixture is $\frac{8}{19}$

12. In the equation, $X + CN^- + H_2O + O_2 \longrightarrow [X(CN)_2]^- + 4OH^-$; 'X' can be
    (A) Au (B) Ag_2S
    (C) Sb (D) As

13. 4.5g of Al (atomic mass = 27 amu) is deposited at cathode from Al^{+3} solution by a certain quantity of charge. The correct statement/s is are:
    (A) The volume of H_2 produced at 0°C and 1atm pressure from H^+ ions in solution by the same quantity of charge will be 5.6 *l*.
    (B) The volume of H_2 produced at 0°C and 1atm pressure from H^+ ions in solution by the same quantity of charge will be 22.4 *l*.
    (C) The number of moles of Al deposited will be 0.166 moles.
    (D) The number of faraday used for reduction of 1 mole of Al^{+3} is $\frac{3}{18}$F.

14. Which of the following statements are true regarding tetrahedral voids in fcc lattice?
    (A) Tetrahedral voids formed by one corner spheres and three face spheres.
    (B) Two tetrahedral voids can exist on one diagonal body line in fcc cubic.
    (C) Tetrahedral voids are equal to number of spheres involved in body.
    (D) Volume of tetrahedral voids are half of the octahedral voids.

15. Which of the following is/are the mixed salts?
    (A) Rochelle's salt (sodium potassium tartrate) (B) Bleaching powder
    (C) Calcium hypochlorite (D) Mohr's salt

Rough work

SECTION – C

(Integer Answer Type)

This section contains **5 questions.** The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.

1. Freezing point of 0.2 M KCN solution is –0.7°C. On adding 0.1 mole of $Hg(CN)_2$ to one litre of the 0.2 M KCN solution, the freezing point of the solution becomes –0.53°C due to the reaction $Hg(CN)_2 + mCN^- \rightarrow Hg(CN)_{m+2}^{m-}$. What is the value of m assuming molality = molarity?

2. In the electrolysis of KI, I_2 is formed at the anode by the reaction;

   $2I^- \longrightarrow I_2 + 2e^-$

   After the passage of current of 0.5 ampere for 9650 seconds, I_2 formed required 40 ml of 0.1 M $Na_2S_2O_3 \cdot 5H_2O$ solution in the reaction;

   $I_2 + 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2I^-$

   What is the current efficiency?

3. What is the oxidation number of Fe in violet coloured complex $Na_4\left[Fe(CN)_5(NO)S\right]$?

4. An alloy of copper, silver and gold is found to have cubic lattice in which Cu atoms constitute CCP. If Ag atoms are located at the edge centres and Au atom is present at body centre, the alloy will have the formula $Cu_xAg_yAu_z$. What is the value of X?

5. If $E^0_{Fe^{2+}/Fe} = 5V$ and $E^0_{Fe^{3+}/Fe^{2+}} = 2V$ then $E^0_{Fe^{3+}/Fe}$ will be.

Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. One Indian, four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian men is seated adjacent to his wife given that each American man is seated adjacent to his wife is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

2. A point P lying inside the curve $y = \sqrt{2ax - x^2}$ is moving such that its shortest distance from the curve at any position is greater than its distance from x-axis. The point P enclose a region whose area is equal to

(A) $\frac{\pi a^2}{2}$ (B) $\frac{a^2}{3}$

(C) $\frac{2a^2}{3}$ (D) $\left(\frac{3\pi - 4}{6}\right)a^2$

3. I write a letter to my friend and don't receive a reply. It is known that one out of m letters does not reach its destination. Then probability that my friend received the letter (it is certain that my friend would have replied, if he did receive my letter)

(A) $\frac{2m-1}{m-1}$ (B) $\frac{m-1}{2m-1}$

(C) $\frac{m+1}{2m+1}$ (D) none of these

4. If $P(\alpha, \beta, \gamma)$ be a vertex of an equilateral triangle PQR where vertex Q and R are $(-1, 0, 1)$ and $(1, 0, -1)$ respectively then P will lie on the plane

(A) $x + y + z + 6 = 0$ (B) $2x + 4y + 32 + 10 = 0$

(C) $x - y + z + 12 = 0$ (D) $x + y + z + 3\sqrt{2} = 0$

Rough work

5. $\vec{a}$ and $\vec{b}$ are non-zero non-collinear vectors such that $|\vec{a}| = 2$, $\vec{a}\cdot\vec{b} = 1$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$. If $\vec{r}$ is any vector satisfying $\vec{r}\cdot\vec{a} = 2$, $\vec{r}\cdot\vec{b} = 8$, $(\vec{r} + 2\vec{a} - 10\vec{b})(\vec{a}\times\vec{b}) = 4\sqrt{3}$ and is equal to $\vec{r} + 2\vec{a} - 10\vec{b} = \lambda(\vec{a}\times\vec{b})$, then λ =

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{1}{4}$ (D) 4

6. If p, q, r are in A.P., then the determinants $\begin{vmatrix} a^2 + 2^{n+1} + 2p & b^2 + 2^{n+2} + 3q & c^2 + p \\ 2^n + p & 2^{n+1} + q & 2q \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix}$ is equal to

(A) 1 (B) 0

(C) $a^2b^2c^2 - 2^n$ (D) $(a^2 + b^2 + c^2) - 2^n q$

7. The area bounded by the curve $y = \frac{3}{|x|}$ and $y + |2 - x| = 2$ is

(A) $\frac{4 - \ln 27}{2}$ (B) $2 - \ln 3$

(C) $2 + \ln 3$ (D) none of these

8. Cards are drawn one by one at random from a well shuffled pack of 52 cards until 2 spades are obtained for the first time. The probability that 20 draws are required for this is

(A) $\frac{{}^{39}C_{18}}{{}^{52}C_{19}} \cdot \frac{{}^{13}C_2}{33}$ (B) $\frac{{}^{39}C_{18} \cdot {}^{13}C_1}{{}^{52}C_{19}} \cdot \frac{12}{33}$

(C) $\frac{{}^{40}C_{19}}{{}^{52}C_{19}} \cdot \frac{12}{13}$ (D) none of these.

9. Let a plane passes through origin and is parallel to line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z+1}{-2}$ such that the distance between plane and line is 5/3, then the equation of plane is

(A) $2x + 2y - z = 0$ (B) $x - 2y - 2z = 0$

(C) $x + 2y + 2z = 0$ (D) $2x - 2y + z = 0$

Rough work

10. Given $q^2 - pr < 0$, $p > 0$, then the value of $\Delta = \begin{vmatrix} p & q & px+qy \\ q & r & qx+ry \\ px+qy & qx+ry & 0 \end{vmatrix}$ is

(A) 0 (B) positive
(C) negative (D) $q^2 + pr$

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. A fair coin is tossed n times. Let x = number of times head occurs. If P(x = 4), P (x = 5) and P (x = 6) are in A.P then the value of n can be
(A) 7 (B) 10
(C) 12 (D) 14

12. If p_1, p_2, p_3 denotes the distance of the plane $2x - 3y + 4z + 2 = 0$ from the planes $2x - 3y + 4z + 6 = 0$, $4x - 6y + 8z + 3 = 0$ and $2x - 3y + 4z - 6 = 0$ respectively, then
(A) $p_1 + 8p_2 - p_3 = 0$ (B) $p_3 = 16p_2$
(C) $8p_2 = p_1$ (D) $p_1 + 2p_2 + 3p_3 = \sqrt{29}$

13. The lines whose vector equations are $\vec{r} = 2\vec{i} - 3\vec{j} + 7\vec{k} + \lambda\left(2\vec{i} + p\vec{j} + 5\vec{k}\right)$ and $\vec{r} = \vec{i} + 2\vec{j} + 3\vec{k} + \mu\left(3\vec{i} - p\vec{j} + p\vec{k}\right)$ are perpendicular for all values of λ and μ, if
(A) $p = -6$ (B) $p = -1$
(C) $p = 1$ (D) $p = 6$

14. The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are of the same length and taken pair wise they form equal angles. If $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$, the coordinates of $\vec{c}$ can be
(A) (1, 0, 1) (B) (-1, 1, 2)
(C) $\left(-\frac{1}{3}, \frac{4}{3}, -\frac{1}{3}\right)$ (D) $\left(\frac{1}{3}, 0, \frac{2}{3}\right)$

Rough work

15. For a, b, c non zero real numbers, $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix} = 0$, then

(A) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$ (B) $\frac{1}{a}+\frac{1}{b\omega}+\frac{1}{c\omega^2}=0$

(C) $\frac{1}{a}+\frac{1}{b\omega^2}+\frac{1}{c\omega}=0$ (D) $\frac{1}{a\omega}+\frac{1}{b\omega^2}+\frac{1}{c}=0$

SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. If the least value of the area bounded by the line y = mx + 1 and the parabola $y = x^2 + 2x - 3$ is α where m is a parameter then the value of $\frac{6\alpha}{32}$ is _____

2. A, B, C and D cut a pack of 52 cards successively in the order given. If the person who cuts a spade first receives Rs. 350 and if the expectation of A is λ then $\left[\frac{\lambda}{64}\right]$ is equal to (where [.] denotes greatest integer function) _____

3. Angle θ is made by line of intersection of planes $\vec{r}\cdot\left(\hat{i}+2\hat{j}+3\hat{k}\right)=0$ and $\vec{r}\cdot\left(3\hat{i}+3\hat{j}+\hat{k}\right)=0$ with $\hat{j}$, where $\cos\theta = \sqrt{\frac{\lambda}{3}}$, then λ is _____

4. The shortest distance between $(1-x)^2 + (x-y)^2 + (y-z)^2 + z^2 = \frac{1}{4}$ and 4x + 2y + 4z + 7 = 0 in 3 – dimensional coordinate system is equal to _____

5. If A′ denotes the transpose of matrix A, A′A = I and |A| = 1 then |A – I| must be equal to (where A is matrix of order 3) _____

Rough work